

Cycle Survival Under Poisson Perturbation of Random Permutations

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Abstract

Let π be a uniformly random permutation of $[n] = \{1, \dots, n\}$, and independently reroute each point i , with probability c/n , to a uniform destination in $[n]$ (leaving $f(i) = \pi(i)$ otherwise). We give a closed form for the $n \rightarrow \infty$ limit of the expected fraction of points left on a cycle of f :

$$\varphi_\infty(c) = \int_0^1 e^{-ct^2} dt = \frac{1}{2} \sqrt{\pi/c} \operatorname{erf}(\sqrt{c}),$$

proved directly on the explicit $n \rightarrow \infty$ limit object (an exploration-process argument with an exact cancellation), together with its entire-function series, a tail asymptotic with a rigorous exponentially-small error bound, and the law of the cyclic mass conditional on exactly K surviving reroutes. That conditional law's mean is proved for every K and coincides, for a structurally different microscopic model, with a theorem of Hansen and Jaworski [5]. The finite- $n \rightarrow \infty$ bridge is proved in full only for $K = 0, 1$ (with an exact $O(1/n^2)$ rate at $K = 1$); for $K \geq 2$ it is reduced, via an unconditionally proved mixing lemma, to a single precisely stated open convergence lemma, which is not resolved here. Every claim below is labeled by its exact status; a companion numerical package cross-checks every closed form against independent brute-force enumeration, Monte Carlo simulation, and quadrature.

1 The model

Fix $n \in \mathbb{N}$ and $c \geq 0$. Let π be a uniformly random permutation of $[n]$. Independently, for each $i \in [n]$ let ξ_i be i.i.d. Bernoulli with $P(\xi_i = 1) = c/n$, and let U_i be i.i.d. uniform on $[n]$, independent of π and of the ξ 's. Define the random mapping

$$f(i) = \begin{cases} U_i & \xi_i = 1 \quad (\text{"}i \text{ is rerouted"} \\ \pi(i) & \xi_i = 0. \end{cases}$$

A point i is *cyclic* for f iff $f^t(i) = i$ for some $t \geq 1$, equivalently iff i lies on a directed cycle of the functional digraph $i \mapsto f(i)$. The observable of interest is

$$\varphi(n, c) := \frac{1}{n} E[\#\{i : i \text{ cyclic for } f\}] = P(1 \text{ is cyclic for } f)$$

(the second equality by exchangeability in i). c is the expected number of rerouted points; $c = 0$ recovers a plain uniform permutation, so $\varphi(n, 0) \equiv 1$.

No name for this exact ensemble was located in the random-mapping literature (see §6): the closest studied family – Jaworski [6], Jaworski [7], Hansen–Jaworski [2, 4, 5] – is parametrized by an in-degree sequence, not by an independently corrupted background permutation.

2 The limit object and the main theorem

The $n \rightarrow \infty$ scaling limit of the cycle structure of a uniform random permutation is the Poisson–Dirichlet object PD(1) [8], with size-biased (“stick-breaking”) representation GEM(1), due to McCloskey (1965) and Patil–Taillie (1977); see Pitman [11], Ch. 3, for the construction and full attribution. Let $L(c)$ be this object equipped with an independent rate- c Poisson process of marks on $[0, 1]$, each carrying an independent uniform destination: an unmarked point advances to the next point on its cycle; a marked point moves instead to its destination. This can be built explicitly from independent $\text{Exp}(1)$ “closure clocks” per arc-head raced against the mark process, with the identification of this construction as the standard exploration representation of PD(1) under lazy revelation cited from [8, 1] and used as the single external structural input; every computation below is self-contained given that one citation. Write $\varphi_\infty(c) := P(x_0 \text{ cyclic in } L(c))$ for a uniform reference point x_0 ; by Fubini and exchangeability this equals $E[\text{Leb}(\text{cyclic set})]$, the natural limit of $\varphi(n, c)$.

Theorem 1 (Closed form). *For every $c \geq 0$,*

$$\varphi_\infty(c) = \int_0^1 e^{-ct^2} dt = \frac{1}{2} \sqrt{\pi/c} \operatorname{erf}(\sqrt{c}),$$

with value 1 at $c = 0$ by continuity.

Proof sketch. Reveal the forward orbit of x_0 step by step; let $t \in [0, 1]$ be the traversed fraction of mass. The orbit decomposes into *arcs* separated by surviving marks. At traversal level t : the π -closure hazard onto a specific open arc-head is $1/(1-t)$ per unit mass (the classical fact that the size-biased cycle length of a uniform permutation is $\text{Unif}(0, 1)$ is the $c = 0$, single-arc-head case of this); a mark at position $s < t$ kills (lands on visited territory) with probability s or survives with probability $1-s$, opening a new arc-head with its own closure clock.

Write $T_0 \sim \text{Unif}(0, 1)$ for x_0 ’s own closure clock (an elementary fact about $\text{Exp}(1)$ primitives), independent of the mark process. Fix $T_0 = t$. For a single surviving mark at $s < t$, the joint probability (over its independent survival Bernoulli and its independent closure clock) that it neither kills *nor* produces a sibling arc-head that closes before t is

$$\underbrace{(1-s)}_{P(\text{survives})} \cdot \underbrace{\frac{1-t}{1-s}}_{P(\text{sibling closes after } t | \text{survives})} = 1-t,$$

independent of s – the survival factor $(1-s)$ exactly cancels the competing-clock tail $(1-t)/(1-s)$. This is the entire mechanism behind the closed form. By the Poisson marking/thinning theorem, the number of “failing” marks in $[0, t]$ is then $\text{Poisson}(ct \cdot t) = \text{Poisson}(ct^2)$ (rate c on an interval of length t , constant fail-probability t , independent of position), so

$$P(x_0 \text{ cyclic} \mid T_0 = t) = P(\text{Poisson}(ct^2) = 0) = e^{-ct^2}.$$

Integrating over $T_0 \sim \text{Unif}(0, 1)$ gives $\varphi_\infty(c) = \int_0^1 e^{-ct^2} dt$; the closed form follows from $u = \sqrt{c}t$ and the definition of erf . $\square$

Remark. A heuristic tail computation that forgets to size-bias “the arc currently being traversed” – treating the inter-mark gaps as unconditional $\text{Exp}(c)$ rather than accounting for the inspection/waiting-time paradox – lands on $\sqrt{\pi/2} c^{-1/2}$ instead of the correct $(\sqrt{\pi}/2) c^{-1/2}$: a factor $\sqrt{2}$ error. The proof above never forms this quantity: the per-mark probability is computed jointly over the pair (survival Bernoulli, closure-clock exponential), never as an averaged arc length, so there is no size-biased object anywhere in the argument to get wrong.

Corollary 1 (Series). $\varphi_\infty(c) = \sum_{k \geq 0} \frac{(-c)^k}{k!(2k+1)}$ for every real c (entire function; proved by the Weierstrass M -test justifying term-by-term integration of e^{-ct^2} 's Taylor series on $[0, 1]$). First terms: $1 - \frac{c}{3} + \frac{c^2}{10} - \frac{c^3}{42} + \dots$; in particular $a_1 = 1/3$.

Corollary 2 (Tail asymptotic, with a rigorous error bound). As $c \rightarrow \infty$,

$$\varphi_\infty(c) = \frac{\sqrt{\pi}}{2} c^{-1/2} - R(c), \quad 0 < R(c) < \frac{e^{-c}}{2c} \quad \forall c > 0,$$

so $\varphi_\infty(c) = \frac{\sqrt{\pi}}{2} c^{-1/2}(1 + O(e^{-c}))$ – a pure power-law tail up to exponentially small corrections. Proved by one integration by parts on the erf tail integral.

3 The conditional- K law and the Hansen–Jaworski connection

Condition on exactly K marks rather than Poisson(c) many.

Lemma 1 (Mean, proved for every K).

$$\varphi_K := \int_0^1 (1 - t^2)^K dt = \frac{4^K (K!)^2}{(2K+1)!}$$

(a Wallis integral). Checks: $\varphi_0 = 1$, $\varphi_1 = 2/3$, $\varphi_2 = 8/15$, $\varphi_3 = 16/35$. Mixing over $K \sim \text{Poisson}(c)$ exactly reproduces Theorem 1: $\sum_K e^{-c} c^K / K! \cdot \varphi_K = \int_0^1 e^{-c} e^{c(1-t^2)} dt = \int_0^1 e^{-ct^2} dt$.

Lemma 2 (Density at $K = 1$, proved). The cyclic-mass random variable $M_1 := \text{Leb}(\text{cyclic set})$ has density $f_{M_1}(x) = 2x$ on $(0, 1)$.

Proof sketch. With a single reroute at a uniform position, the size-biased struck cycle has length $L \sim \text{Unif}(0, 1)$ (classical) and a uniform destination u . If $u \notin$ the struck cycle (probability $1 - L$), the cyclic mass is deterministically $1 - L$; if u lands in the struck cycle (probability L), the cyclic mass is $1 - L + D$ with $D \mid L \sim \text{Unif}(0, L)$. Summing the two branches' contributions to the density of M_1 by the standard mixture change-of-variables formula gives $x + x = 2x$ on $(0, 1)$. $\square$

Conjecture 1 (General- K density, **not proved here**). $f_{M_K}(x) = 2Kx(1 - x^2)^{K-1}$ on $(0, 1)$, for $K \geq 2$.

This reduces to Lemma 2 at $K = 1$; its mean matches φ_K exactly (proved, by integration by parts – necessary but not sufficient for the density itself to be correct); it is supported by Kolmogorov–Smirnov tests at $K = 1, 2, 3$ (no rejection, two independent implementations) and by an external theorem for a structurally different microscopic model with the same conditional- K limit:

Hansen & Jaworski [5], Theorem 7(ii). For their model $\hat{T}_n^r$ – a uniform random mapping with r vertices constrained to in-degree ≤ 1 and $a = n - r$ vertices allowed in-degree ≤ 2 – fixed a and $k = \lfloor xn \rfloor$: $\Pr\{\hat{X}_n^r = k\} \sim \frac{1}{n} \cdot 2ax(1 - x^2)^{a-1}$.

Identical functional form, with $a \leftrightarrow K$; the two microscopic models are structurally unrelated (their model is a uniform draw under an in-degree constraint, with no permutation-plus-reroute mechanism). This coincidence of the conditional- K limit law is external supporting evidence, not a proof, of Conjecture 1 for this ensemble. Literature priority for this specific citation: *already known* – see §6.

4 The finite- n bridge

Theorem 1 and Lemma 1 concern $L(c)$ itself, taken as given. Whether $\varphi(n, c) \rightarrow \varphi_\infty(c)$ as $n \rightarrow \infty$ is a separate question, only partially resolved.

For every finite n , writing $\varphi_n^{(K)}$ for $\varphi(n, c)$ conditioned on exactly K rerouted points (a purely combinatorial quantity, independent of c , by exchangeability),

$$\varphi(n, c) = \sum_{K=0}^n \binom{n}{K} \left(\frac{c}{n}\right)^K \left(1 - \frac{c}{n}\right)^{n-K} \varphi_n^{(K)} \quad (1)$$

is an *exact* identity, not an approximation.

Proposition 1 (Mixing reduction, proved unconditionally). *If $\varphi_n^{(K)} \rightarrow \varphi_K$ for every fixed $K \geq 0$, then $\varphi(n, c) \rightarrow \varphi_\infty(c)$.*

Proof sketch. Split $\varphi(n, c) - \varphi_\infty(c)$ into a term measuring $\text{Binomial}(n, c/n)$ vs. $\text{Poisson}(c)$ total-variation distance (which $\rightarrow 0$ by Scheffé’s lemma [12] applied to the elementary pointwise Poisson-limit computation – a quantitative rate $d_{TV} \leq c^2/n$ is also available via Le Cam’s bound [10], though only the qualitative limit is needed here) and a term measuring $\varphi_n^{(K)} - \varphi_K$ averaged over K , controlled uniformly in n by a Chernoff-type tail bound on $\text{Binomial}(n, c/n)$ derived from scratch. Both vanish as $n \rightarrow \infty$. $\square$

$K = 0$ is trivial and exact for every n ($f = \pi$, a bijection, every point cyclic: $\varphi_n^{(0)} = 1$).

Proposition 2 ($K = 1$ bridge, proved exactly). *For every $n \geq 1$, $\varphi_n^{(1)} = \frac{2}{3} + \frac{1}{3n^2}$, exactly.*

Proof sketch. Fix the rerouted index; π is then an unconditioned uniform permutation, and the cycle C containing it has length L , exactly uniform on $\{1, \dots, n\}$ (a classical exact combinatorial fact, not merely a limit). Points outside C remain cyclic regardless of the reroute’s target U . A case split on U inside C (missing C entirely; landing on the reroute source itself; landing d steps into C) gives an exact count of cyclic points within C given $L = \ell$; averaging over $L \sim \text{Unif}\{1, \dots, n\}$ and U gives the stated closed form. $\square$

Open Lemma (Fixed- K bridge, $K \geq 2$). $\lim_{n \rightarrow \infty} \varphi_n^{(K)} = \varphi_K$ for every fixed integer $K \geq 2$. **Neither proved nor disproved here.**

A single reroute disturbs exactly one background cycle (tractable, the mechanism behind Proposition 2); $K \geq 2$ reroutes can strike the same cycle in either order, or strike different cycles whose severed pieces later re-link through a further reroute’s target – a combinatorial case analysis not resolved here. Exact enumeration at $K = 2$ for n up to 8 shows $\varphi_n^{(2)}$ decreasing monotonically toward $\varphi_2 = 8/15$, but the rescaled deviation $n^2(\varphi_n^{(2)} - \varphi_2)$ is *not* settling to a constant over the tested range – so, unlike $K = 1$ ’s clean $O(1/n^2)$ rate, no specific rate for $K \geq 2$ should be assumed (§5).

Proposition 3 (Conditional convergence). *For every fixed $c \geq 0$, $\varphi(n, c) \rightarrow \varphi_\infty(c)$ as $n \rightarrow \infty$, conditional on Open Lemma 4 holding for every $K \geq 2$.*

This follows immediately from Proposition 1 plus Proposition 2 and the trivial $K = 0$ case, given the open lemma. It is stated as a conditional proposition, not a theorem, because the open lemma is exactly what is missing; the empirical control that exists (§5) is evidence for, not a substitute for, resolving it.

5 Numerics

A companion package (`tamesis-cycle-survival/`, clean-room implementation, `numpy/scipy` only, fixed seeds) cross-checks every closed form above:

- **Series vs. closed form vs. independent quadrature** (`simulations/asymptotics.py`): agree to $< 10^{-9}$ across $c \in [0.01, 10]$ for the series (limited by double-precision catastrophic cancellation of the alternating series beyond $c \sim 15$, a floating-point artifact, not a defect of the series itself) and to $< 10^{-8}$ for erf-vs-quadrature up to $c = 500$.
- **Tail bound** (Corollary 2): the rigorous bound $0 < R(c) < e^{-c}/(2c)$ holds at every tested $c \in \{1, 2, 5, 10, 25, 50, 100, 500\}$ (at $c \geq 50$ the bound is below double-precision resolution for $\varphi_\infty(c)$'s magnitude and $R(c)$ correctly rounds to zero).
- **Exact finite- n enumeration** (`simulations/finite_n.py`, no sampling): Proposition 2's closed form $\varphi_n^{(1)} = 2/3 + 1/(3n^2)$ verified as an exact rational identity by brute-force enumeration for $n = 1, \dots, 9$; Open Lemma 4 explored at $K = 2$ for $n = 2, \dots, 8$, confirming monotone convergence toward $\varphi_2 = 8/15$ without a settled rate; the full exact mixture (1) for $n = 2, \dots, 4$ shows $|\varphi(n, c) - \varphi_\infty(c)|$ shrinking monotonically in n at fixed $c \in \{0.5, 1, 2\}$.
- **Monte Carlo at large n** (`simulations/monte_carlo.py`, $n = 20000$, 300 trials/cell, seed 20260822): all tested $c \in \{0.1, 0.5, 1, 2, 5, 10, 30\}$ within $|z| < 4$ of $\varphi_\infty(c)$.
- **Poisson-mixture consistency**: $\sum_K \text{Poisson}(K; c) \varphi_K$ reproduces $\varphi_\infty(c)$ to $< 10^{-8}$ for every tested c , confirming Lemma 1's consistency remark numerically.

A pytest suite of 49 checks (`tests/test_formula.py`) runs all of the above at small scale in under two seconds; see the package `README.md` for exact reproduction commands for every number in this paper.

6 Literature priority

A dedicated search (22 queries) traced the closest published family – Jaworski [6, 7], Hansen–Jaworski [2, 3, 4, 5] – and reached a verdict per component of this result:

- *The model* (§1): not found under this construction; the closest family is parametrized by in-degree sequence, not permutation-plus-independent-reroute.
- *The limit object* (PD(1) + Poisson marks): each ingredient is classical; the specific combination is not built this way in the source literature, which never starts from a permutation.
- *The conditional- K law* (§3): **already known** – Hansen & Jaworski [5], Theorem 7(ii), for a structurally different model.
- *The Poisson-mixture closed form* (Theorem 1): not found; the strongest candidate component of this result for being previously unpublished. The mixture is a question that arises naturally only from the permutation-plus-reroute side (where $\#\text{reroutes} \sim \text{Binomial}(n, c/n) \rightarrow \text{Poisson}(c)$ is immediate), so its absence from the in-degree-sequence literature is consistent with, not independent evidence against, its being previously unpublished.

- *The tail exponent* $1/2$: universal across the whole random-mapping literature (e.g. the classical $\#\text{cyclic points} \sim \sqrt{\pi n/2}$ for a uniformly random mapping); the exact coefficient $\sqrt{\pi}/2$ and the purely exponential correction structure of Corollary 2 were not found.

This search is broad but not exhaustive: one classical source (Kolchin, *Random Mappings* [9]) remains unverified for lack of full-text access, and is reported here as inaccessible, not as ruled out. Every claim of priority in this paper is qualified as “not found in the searches performed,” never as “does not exist” or “first.”

7 Open problems

1. **Open Lemma 4** itself: the fixed- K finite- n bridge for $K \geq 2$. This is the single hypothesis separating Proposition 3 from a full theorem.
2. **The rate** of that bridge, once resolved: $K = 0$ is exact (rate 0), $K = 1$ is exact at rate $1/(3n^2)$; $K = 2$'s rescaled deviation is not observed to settle over $n \leq 8$ (§5). No specific rate for $K \geq 2$ should be assumed without further work.
3. **Conjecture 1** (general- K density) and its Poisson-mixture consequence, the full unconditional law $M(c) \stackrel{d}{=} \min(1, \sqrt{E/c})$, $E \sim \text{Exp}(1)$.
4. A locally-uniform-in- c (not merely pointwise) version of Proposition 3.
5. Any second-moment, concentration, or CLT statement for the cyclic fraction of either $\varphi(n, c)$ or $L(c)$ – entirely untouched here.
6. A first-principles (non-cited) proof of the exploration-process representation of PD(1) used as the single external input in §2.

References

- [1] Richard Arratia, A. D. Barbour, and Simon Tavaré. *Logarithmic Combinatorial Structures: A Probabilistic Approach*. European Mathematical Society, 2003.
- [2] Jennie C. Hansen and Jerzy Jaworski. Compound random mappings. *Journal of Applied Probability*, 39(4):712, 2002.
- [3] Jennie C. Hansen and Jerzy Jaworski. A new random mapping model, 2006.
- [4] Jennie C. Hansen and Jerzy Jaworski. Predecessors and successors in random mappings with exchangeable in-degrees. *Random Structures & Algorithms*, 33(1):105–126, 2008.
- [5] Jennie C. Hansen and Jerzy Jaworski. Structural transition in random mappings. *Electronic Journal of Combinatorics*, 21(1):#P1.18, 2014.
- [6] Jerzy Jaworski. On a random mapping $(t, \mathbb{P}_j)$. *Journal of Applied Probability*, 21(1):186–191, 1984.
- [7] Jerzy Jaworski. Predecessors in a random mapping. *Random Structures & Algorithms*, 13, 1998.

- [8] J. F. C. Kingman. Random discrete distributions. *Journal of the Royal Statistical Society, Series B*, 37, 1975.
- [9] V. F. Kolchin. *Random Mappings*. Optimization Software / Springer, 1986. Translation of *Sluchainye otobrazheniya*. Full text not accessible during the priority search behind this paper; reported as inaccessible, not as ruled out as a possible antecedent of the model in §1.
- [10] Lucien Le Cam. An approximation theorem for the poisson binomial distribution. *Pacific Journal of Mathematics*, 10:1181–1197, 1960.
- [11] Jim Pitman. Combinatorial stochastic processes. In *École d’Été de Probabilités de Saint-Flour XXXII – 2002*, volume 1875 of *Lecture Notes in Mathematics*. Springer, 2002.
- [12] Henry Scheffé. A useful convergence theorem for probability distributions. *Annals of Mathematical Statistics*, 18:434–438, 1947.